Hamid Ettayyebi

Data Scientist

Portfolio: hamid701.github.io
github.com/Hamid701
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PROFESSIONAL SUMMARY

Data Scientist with 5+ years of research-to-production experience specializing in time series forecasting and ML engineering. Expert in developing reproducible ML solutions following best practices, from statistical modeling to production deployment.

SKILLS

- **ML/AI:** Time Series, Deep Learning, Regression, Classification and Statistical Modeling, Anomaly / Fault Detection
- **Programming & Data:** Python (Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, SciPy), SQL (PostgreSQL, MySQL).
- **MLOps & Deployment:** FastAPI, Docker, CI/CD, Git workflows, pytest, reproducible pipelines
- **Visualization:** Matplotlib, Seaborn, Plotly, Streamlit, Dashboards
- **Languages:** Arabic (Native), English (C1), French (B2), Italian (A2: currently learning)

PROFESSIONAL EXPERIENCE

Data Scientist **Jan 2016–present**
Hermes Corporate *Florence, Italy*

- Built and deployed an end-to-end predictive-maintenance analytic for gas-turbine lube-oil leak detection, combining wavelet denoising with multi-window rolling-regression slope detection on 10-min sensor streams to cut time-to-detection from 13h to under 2h; deployed across a 1,000-turbine fleet, projected to avoid €1M+/year in downtime and equipment damage.
- Engineered multiple independent false-positive gates, cutting false leak alerts to a single confirmed event across a 2-year multi-asset validation while preserving every true detection.
- Architected a modular, production-grade pipeline with per-machine self-calibrating thresholds and cross-batch state persisted in PostgreSQL, hardened by a 255-test regression suite, integrating the new detector with a legacy model into one analytic.

PhD Researcher: Applied Data Scientist **Jan 2017–Sep 2021**
Mohammed V University *Rabat, Morocco*

- Analyzed complex business requirements for solar energy forecasting systems, determining ML solution feasibility and developing time series forecasting models that reduced forecast error by 15%, improving grid reliability for utility partners.
- Built production-ready ML solutions incorporating software engineering best practices, implementing automated data pipelines, and model validation frameworks that cut data preparation time by 30% and accelerated testing cycles.
- Designed and optimized ANNs through systematic experimentation, outperforming statistical baseline models by 13.43% while collaborating with engineers to monitor performance in operational environments, enabling more accurate capacity planning and reducing reserve requirements by an estimated 5%.

Mathematics Teacher **Sep 2017–Aug 2024**
Regional Academy for Training and Education *Rabat, Morocco*

- Taught 1,120+ students (pass rate 82% → 92%) and trained 5 peer tutors, improving first-attempt problem-solving success by 25

PROJECTS

Rentelligence AI: Dynamic Pricing Engine **Jan 2025–Apr 2025**
End-to-end project - GitHub | Live demo *Rome, Italy*

- Deployed an XGBoost-powered pricing engine in production, enabling on-demand rental price recommendations in minutes instead of days and generating an estimated €50K in annual productivity savings for property managers.
- Engineered 15+ geospatial features from 12,000+ listings and raised data reliability from 55% to 95%, saving 20 manual hours/week.

Solar Forecasting Pipeline **Aug 2024–Jan 2025**
End-to-end project - GitHub | Live demo *Rome, Italy*

- Built and deployed a Transformer-based solar forecasting pipeline trained on 10+ years of meteorological data, reducing forecast error by 20% and cutting annual grid misallocation costs by an estimated €200K.
- Built and deployed a Transformer-based solar forecasting pipeline on 10+ years of meteorological data, reducing forecast error by 20% (€200K/year in avoided grid misallocation), with FastAPI/Plotly dashboards that cut decision lag by 50%.

EDUCATION & CERTIFICATIONS

Master's Degree in Mathematics and Applications, Statistics, and Numerical Computing
Mohammed V University *Rabat, Morocco*
IBM Data Science Specialization **Feb 2025**
Credential ID: W85E3XU7YR5X

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